OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

14.3 Signal dependency

Signal dependency means that one signal controls the output of another signal.

Elements in UML model

`<dependency>` element

In ASAM OpenDRIVE, signal dependency is represented by the `<dependency>` element within the `<signal>` element.

```
UML class: t_road_signals_signal_dependency
XML tag:   <dependency> (Multiplicity: 0..*)
```

Signal dependencies limit or extend the validity of one signal by an additional signal. For example, a speed limit sign of 60 km/h that is only valid for trucks, specified by a supplementary sign. One signal may have multiple dependencies.

Table 122. Attributes of the <dependency> element

Name	Type	Use	Description
`id`	string	required	ID of the controlling signal
`type`	string	optional	Type of the dependency, Free text, depending on application

XML example

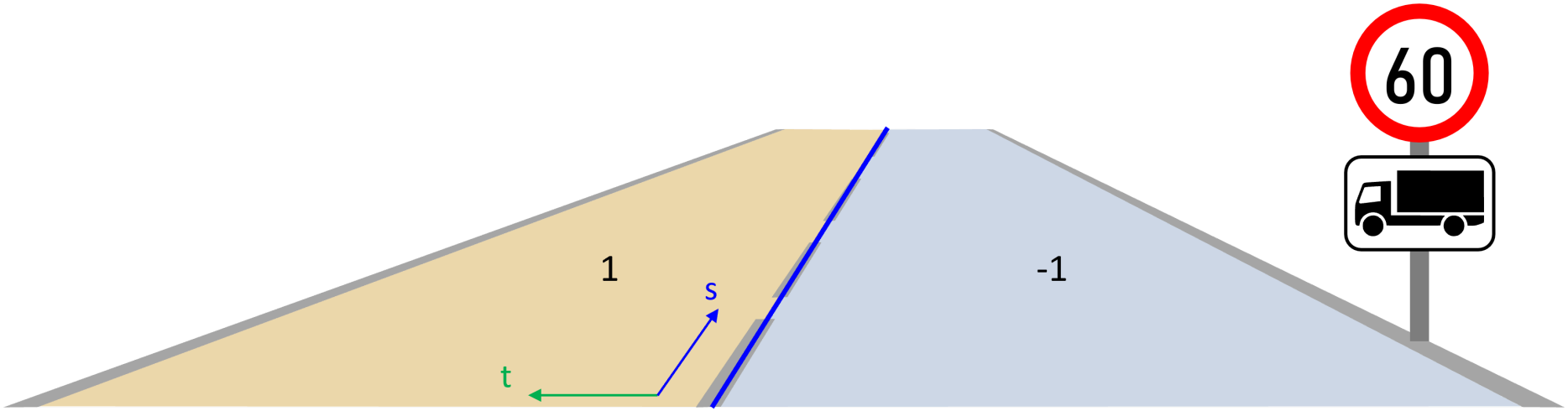

Figure 129. Lane and type specific speed limit

Figure 129 shows the dependency between the speed limit signal and the signal to make the lane valid for specific traffic participants only.

```
<signals>
    <signal s="50.0"
            t="-4.0"
            id="1"
            name="SpeedLimit60"
            dynamic="no"
            orientation="+"
            zOffset="1.90"
            type="274"
            country="DE"
            countryRevision="2013"
            subtype="56"
            value="60.0"
            unit="km/h"
            hOffset="0.0 "
            pitch="0.0"
            roll="0.0"
            height="0.61"
            width="0.61">
        <dependency id="2"/>
    </signal>
    <signal s="50.0"
            t="-4.0"
            id="2"
            name="LorriesOnly"
            dynamic="no"
            orientation="+"
            zOffset="1.56"
            type="1048"
            country="DE"
            countryRevision="2013"
            subtype="12"
            hOffset="0.0"
            pitch="0.0"
            roll="0.0"
            height="0.33"
```

```
            width="0.60">
    </signal>
</signals>
```

Rules

The following rules apply to dependency elements:

- asam.net:xodr:1.7.0:road.signal.dependency.multiple_dependency: A signal may have multiple dependencies.
- The type of dependency is not specifically defined in ASAM OpenDRIVE and may be set in the application.

Rules regarding the type of dependency are defined in the application and are not stored in ASAM OpenDRIVE.

Related topics

- Section 14.1, "Introduction to signals"
- Section 14.4, "Signal reference"